The empirical recurrence for OEIS A254771, recovered from its tail

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Abstract

OEIS A254771 records "Empirical recurrence of order 69" and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 1017$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 13$, so the recurrence is proved for every $n \geq 82$. Its last coefficient is $c_{69} = 147456 \neq 0$, so no further tail satisfies anything shorter; any order-69 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry's recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A254771 is "Number of (n+2)X(4+2) 0..1 arrays with every 3X3 subblock sum of the two maximums of the central row and the central column and the two medians of the diagonal and antidiagonal nondecreasing horizontally, vertically and ne-to-sw antidiagonally.". It has offset 1 and begins

$$48168, 207006, 1667480, 14533460, 113507484, 914580497, 7597019270, 62541963004, \ \ldots$$

The entry states:

> Empirical recurrence of order 69 (see link above)

As of the "Last modified" line on the live entry (Jul 23 2025 14:55:25, revision 6) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry's condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u, and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) \ = \ \iota^{\top} M^{\,j(n)} \tau$$

for a linear reindexing j fixed by the entry's shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 1017$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 1017$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \ldots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a. □

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 69: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 13$ is the first whose minimal order is exactly the stated 69.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 2034$ exact terms from $n = 13$ on, it returns order 69 — the order the entry states — with integer coefficients:

$$a(n) \;=\; \sum_{i=1}^{69} c_i \, a(n-i),$$

c_1	7	c_{19}	797651581	c_{37}	35460952656	c_{55}	−12785780224
c_2	36	c_{20}	196764022	c_{38}	−177360658450	c_{56}	19899420928
c_3	−153	c_{21}	218351793	c_{39}	−17915841956	c_{57}	25710865408
c_4	−400	c_{22}	−1338029087	c_{40}	65925839760	c_{58}	156653056
c_5	−2266	c_{23}	−3723551276	c_{41}	42791267864	c_{59}	−8957111296
c_6	7759	c_{24}	217851777	c_{42}	43814973360	c_{60}	−4370827264
c_7	50483	c_{25}	1071034411	c_{43}	32563312416	c_{61}	−1493762048
c_8	−68824	c_{26}	6544550550	c_{44}	−113793047968	c_{62}	1569521664
c_9	−169897	c_{27}	4470553766	c_{45}	−56156543936	c_{63}	705497088
c_{10}	13829	c_{28}	3307537241	c_{46}	48982549936	c_{64}	1773568
c_{11}	−1413576	c_{29}	834844009	c_{47}	50523765376	c_{65}	−24756224
c_{12}	121237	c_{30}	−27564165998	c_{48}	20269561536	c_{66}	−52477952
c_{13}	12603668	c_{31}	16499130454	c_{49}	−52998880896	c_{67}	6750208
c_{14}	16662226	c_{32}	−38611014867	c_{50}	43745928704	c_{68}	−606208
c_{15}	−17086887	c_{33}	−25029705372	c_{51}	−26589974400	c_{69}	147456
c_{16}	−93068217	c_{34}	110576779318	c_{52}	−2863970048		
c_{17}	−208974569	c_{35}	−35575951604	c_{53}	29812727936		
c_{18}	157418025	c_{36}	49913733728	c_{54}	−49249726976		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 82$, and it is the recurrence OEIS A254771 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 13$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{69} - c_1 x^{69-1} - \cdots - c_{69}$. Its constant term is $-c_{69} = -147456$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 69 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 69, so $\deg q = 69$, and it is monic. It holds from some index t on. From $\max(t, 13)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 69, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 15 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry's English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 69, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry's.

Third, the recovered recurrence was evaluated directly on the entry's own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is 147456.

Remark 4. The conjecture's own statement was never read. The entry pins it down to the family of order-69 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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